

Joins in SQL

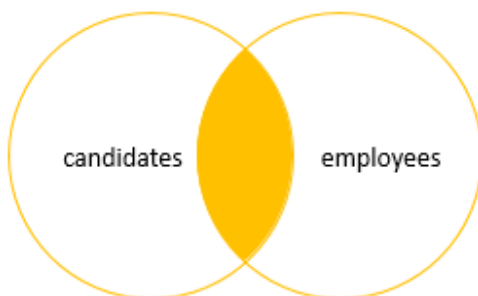
Joins are used to relate one or more tables in SQL Server. Joins are a part of a SQL Statement that retrieves rows from a table or tables according to specified conditions.

Types of Joins:

- Inner join
- Left outer join
- Right outer join
- Full outer join
- Cross join
- Self Join

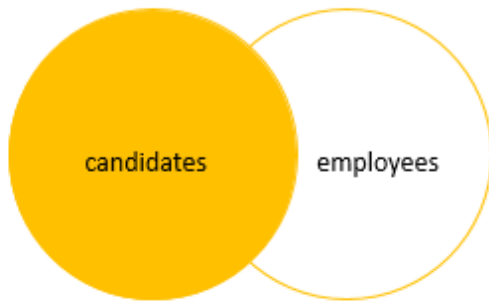
1. **Inner Join:** Inner join produces a data set that includes rows from the left table, matching rows from the right table.

The following Venn diagram illustrates the result of the inner join of two result sets:

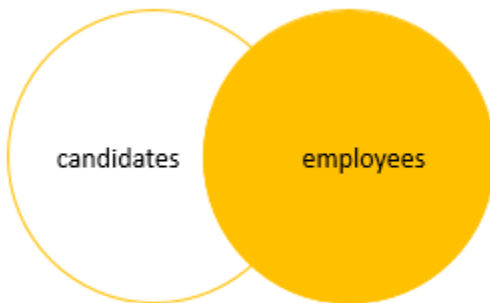


2. **Left outer join:** Left join selects data starting from the left table and matching rows in the right table. The left join returns all rows from the left table and the matching rows from the right table. If a row in the left table does not have a matching row in the right table, the columns of the right table will have nulls.

The following Venn diagram illustrates the result of the left join of two result sets:

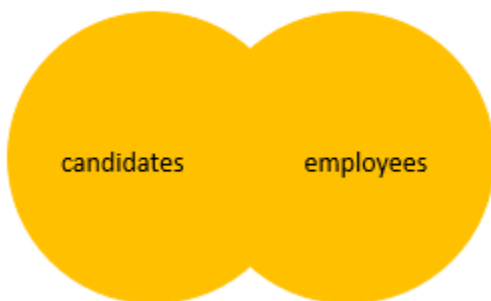


3. **Right outer join:** The right join or right outer join selects data starting from the right table. It is a reversed version of the left join.



4. **Full outer join:** The full outer join or full join returns a result set that contains all rows from both left and right tables, with the matching rows from both sides where available. In case there is no match, the missing side will have NULL values.

The Venn diagram that illustrates the full outer join:



5. **Cross join:** The CROSS JOIN joined every row from the first table (T1) with every row from the second table (T2). In other words, the cross join returns a Cartesian product of rows from both tables.

- 6. Self Join:** A self join allows you to join a table to itself. It helps query hierarchical data or compare rows within the same table.

A self join uses the inner join or left join clause. Because the query that uses the self join references the same table, the table alias is used to assign different names to the same table within the query.